Taufiqul Islam Khan

Sessional Instructor • Research Assistant • MAMBA Lab • University of Manitoba
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CAREER OBJECTIVE

Educator and researcher with over six years of experience in university teaching and software engineering. I enjoy helping students learn, and my research is in software engineering, code review automation, and AI tools for developers.

WORK EXPERIENCE

University of Manitoba — *January 2026 – Ongoing*
Sessional Instructor — *Winnipeg, Manitoba, Canada*

- Teach COMP 3350 Software Engineering I (Winter 2026) and COMP 2450 Software Development 1 (Fall 2026) to undergraduate students
- Teach agile methods, software design principles, and software testing
- Help students with course material and give clear feedback on their work
- Guide student teams through term-long software projects; write and grade assignments and exams
- Manage course material on UMLearn, hold regular office hours, and invigilate exams

BRAC University — *June 2021 – December 2024*
Full-Time Lecturer — *Dhaka, Bangladesh*

- Course Coordinator for Software Engineering and Introduction to Robotics (Lab): led the faculty team teaching each course and set the syllabus and exams for all sections
- Taught Software Engineering, Data Structures, Programming Language I, Computer Interfacing, and Introduction to Robotics
- Ran lab sessions and did department administrative work alongside teaching
- Prepared exam papers, graded assessments, and invigilated theory and lab exams

BJIT Ltd. — *October 2021 – July 2022*
Software Engineer — *Dhaka, Bangladesh*

- Built and maintained backend services using Spring Boot, REST APIs, and the 3dSpace database
- Used Jira, Redmine, Bitbucket, and GitHub to track tasks and manage code in an agile team
- Used design patterns (Adapter, Observer), wrote unit tests, and refactored code to make it easier to maintain

ServiceEngine BPO — *July 2020 – October 2021*
Software Engineer — *Dhaka, Bangladesh*

- Built web scrapers in Groovy, Python, and Selenium to collect data from websites at scale
- Converted data from PDFs and images into structured formats using OCR and image processing
- Built and deployed supervised machine learning models in Python and TensorFlow to check the quality of processed data
- Set up low-cost automated data processing pipelines with Jenkins CI/CD and AWS

Department of CSE, BRAC University — *January 2018 – May 2020*
Student Tutor — *Dhaka, Bangladesh*

- Helped instructors teach Algorithms and Digital Logic Design to undergraduate students
- Helped run lab sessions, invigilated exams, and helped with marking and student feedback

EDUCATION

University of Manitoba — *Winnipeg, Canada*
Ph.D. in Computer Science (Admitted) — *Starting January 2027*
M.Sc. in Computer Science (Current Student) — *2025 – 2026*

- Research Assistant at MAMBA Lab under Professor Shaowei Wang, working on code review automation and LLM-based developer tools

BRAC University — *2020*
B.Sc. in Computer Science & Engineering | CGPA: 3.9 / 4.0 — *Dhaka, Bangladesh*

- Thesis: 6LoWPAN Based Futuristic Architecture for Smart Home

Dhaka College — *2015*
H.S.C. in Science | GPA: 5.00 / 5.00 — *Dhaka, Bangladesh*

- Achieved Golden GPA 5 (highest possible result in Bangladesh) and placed 82nd in Dhaka Board

Shere Bangla Nagar Govt. Boys' High School — *2013*
S.S.C. in Science | GPA: 5.00 / 5.00 — *Dhaka, Bangladesh*

- Achieved Golden GPA 5 (highest possible result in Bangladesh)

ACHIEVEMENTS

- Fully Funded M.Sc. Research Assistantship, University of Manitoba (2025–Present)
- Named on the VC's List and Dean's List, BRAC University (2016–2020)
- Merit-based Scholarship, 70% tuition fee waiver, BRAC University (2016–2020)
- Higher Secondary Government Scholarship, Placed 82nd in Dhaka Board, Dhaka College (2015)
- Primary School Government Scholarship, Placed 17th in District, Manikganj, Dhaka (2007)

TALKS, PRESENTATIONS & COMMUNITY ENGAGEMENT

Poster Presentation, CSER (Consortium for Software Engineering Research), Spring 2026

- Presented my survey on code review benchmarks in the LLM era and talked with researchers from Canadian universities and industry about what current evaluation methods are missing

Invited Talk: "From Campus to Canada", BRAC University Computer Club

- Talked about my path from undergraduate studies to teaching to graduate school in Canada, including how to get into research, what industry expects, and how to apply to Canadian universities

Freshers' Orientation Sessions & Workshops, Department of CSE, BRAC University

- Ran several orientation and mentoring sessions for new CS students, and a hands-on workshop for first- and second-year students on finishing their first coding project

Community Talks, Local Schools & Coaching Centers, Dhaka

- Spoke to secondary school students at local schools and coaching centers about computer science careers, problem solving, and preparing for SSC/HSC exams

SKILLS

Frameworks:	Spring Boot, React, Flask, Node.js, Express, TensorFlow, Selenium
Programming:	Python, Java, JavaScript, SQL, HTML, CSS, Groovy, R
Version Control:	Git, GitHub, GitLab, Bitbucket, Jira, Docker, Postman
Miscellaneous:	Linux, Shell (Bash), LaTeX (Overleaf), Microsoft Office, MongoDB
Research:	Software Engineering, Code Review Automation, LLM/NLP, Data Science, Machine Learning, IoT
Soft Skills:	Taking Initiative, Time Management, Teamwork, Problem Solving, Documentation, Public Speaking

PROJECTS

Leveraging Blockchain and Machine Learning for a Secure and Transparent Voting System (University of Manitoba, 2025)

- Built a full-stack e-voting system that uses blockchain smart contracts and ML-based face recognition to keep votes secure, tamper-proof, and private | Blockchain, Smart Contracts, Machine Learning, Facial Recognition

Smart Contract for a Decentralized Energy Marketplace & Auction Mechanisms (University of Manitoba, 2025)

- Wrote Solidity smart contracts for a decentralized energy marketplace that supports English and Vickrey (second-price) auctions | Solidity, Remix IDE, Blockchain

Enterprise Backend Services with Spring Boot (BJIT Group, 2022)

- Built backend services for a Japanese client's product lifecycle management system; used Adapter and Observer design patterns to keep the code modular | Spring Boot, REST API, Java, 3dSpace

Number Plate Recognition System (ServiceEngine BPO, 2021)

- Built a real-time computer vision system that detects and reads license plates from traffic video; used in production | Python, TensorFlow, OpenCV, OCR

Automated Data Pipelines (ServiceEngine BPO, 2021)

- Built automated data scraping and processing pipelines on AWS with Jenkins CI/CD, which replaced a large amount of manual data work | Python, Groovy, Selenium, AWS, Jenkins

Opu's Coaching Android App (BRAC University, 2019)

- Built an Android app that connects teachers and students for online classes, with session scheduling and real-time sync through Firebase | Android, Java, Firebase

PUBLICATIONS

A Survey of Code Review Benchmarks and Evaluation Practices in Pre-LLM and LLM Era
TI Khan, S Wang, H Zhang, TH Chen • arXiv Preprint • https://doi.org/10.48550/arXiv.2602.13377 • 2026

6LoWPAN Based Futuristic Smart Architecture for Home Automation
*SMJ Hasan, M Ali, TI Khan, F Afrin, MM Islam • ICAICT 2020 • **DOI:** 10.1109/ICAICT51780.2020.9333483 • Cited by: 3*

Transfer Learning Using Convolutional Neural Networks for COVID-19 Detection with X-Ray Images
PK Das, M Rashidunnabi, TI Khan, SRR Ahmad • scholar.google.ca/citations?user=yINPGNUAAAAJ

REFEREES

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